

ABSTRACT

The Project presents the overall design of home automation system with low cost and wireless system, enhance convenience, safety, and energy efficiency. Leveraging interconnected devices and sensors, the system enables remote monitoring and control of various home appliances and environmental parameters. Key components include Node MCU (ESP8266) for connectivity, relay modules for appliance control, PIR sensor for occupancy detection, ultrasonic sensor for distance measurement, flame sensor for fire detection, and MQ2 sensor for gas monitoring. Through a user-friendly interface accessible via smartphones or web platforms like Blynk app, users can remotely manage their home environment, receiving real-time updates and alerts. By integrating IoT technologies, this system aims to streamline daily routines, improve security, and reduce energy consumption, ultimately creating a more comfortable and sustainable living space.

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CHAPTER-1

INTRODUCTION

In This Project, You Learn Full Home Automation using Blynk App & Node MCU (esp8266). This Project is the Most Common & Popular Project Across the Internet. Home Automation Mean we controlled Home Light, Measuring the Gas Level, Measuring Temperature & humidity And Measuring the Water Tank Level.

In This Project, we are measuring the Parameter and Controlling the two applications. all possibilities will be done by WIFI Through I mean the NODE MCU is inbuilt WIFI Controller. And when the WIFI is Connected to the NODE MCU we control the applications and measuring the Parameter Anywhere in the World. All data will Show in Blynk2.0 App.

Home automation refers to the integration of technology and systems within residence to provide convenience, comfort, energy efficiency, and security. Through interconnected devices and smart technologies, homeowners can remotely control various aspects of their homes, such as lighting, heating, cooling, security cameras, and entertainment systems, often via smartphones or voice commands. This integration offers not only convenience but also potential cost savings and increased safety, making home automation an increasingly popular trend in modern households.

CHAPTER-2

HOME AUTOMATION SYSTEM

2.1 WHAT IS HOME AUTOMATION?

“Home automation” refers to the automatic and electronic control of household features, activity, and appliances. In simple terms, it means you can easily control the utilities and features of your home via the Internet to make life more convenient and secure, and even spend less on household bills. Read on to find answers to some of the most common questions about home automation technology, and get a few ideas for home automation solutions to incorporate in your home

2.2 HOW DOES IT WORKS

Home automation is a network of hardware, communication, and electronic interfaces that work to integrate everyday devices with one another via the Internet. Each device has sensors and is connected through WiFi, so you can manage them from your smartphone or tablet whether you’re at home, or miles away. This allows you to turn on the lights, lock the front door, or even turn down the heat, no matter where you are

There are three main elements of a home automation system: sensors, controllers, and actuators.

Sensors can monitor changes in daylight, temperature, or motion detection. Home automation systems can then adjust those settings (and more) to your preferences

Controllers refer to the devices — personal computers, tablets or smartphones — used to send and receive messages about the status of automated features in your home

Actuators may be light switches, motors, or motorized valves that control the actual mechanism, or function, of a home automation system. They are programmed to be activated by a remote command from a controller.

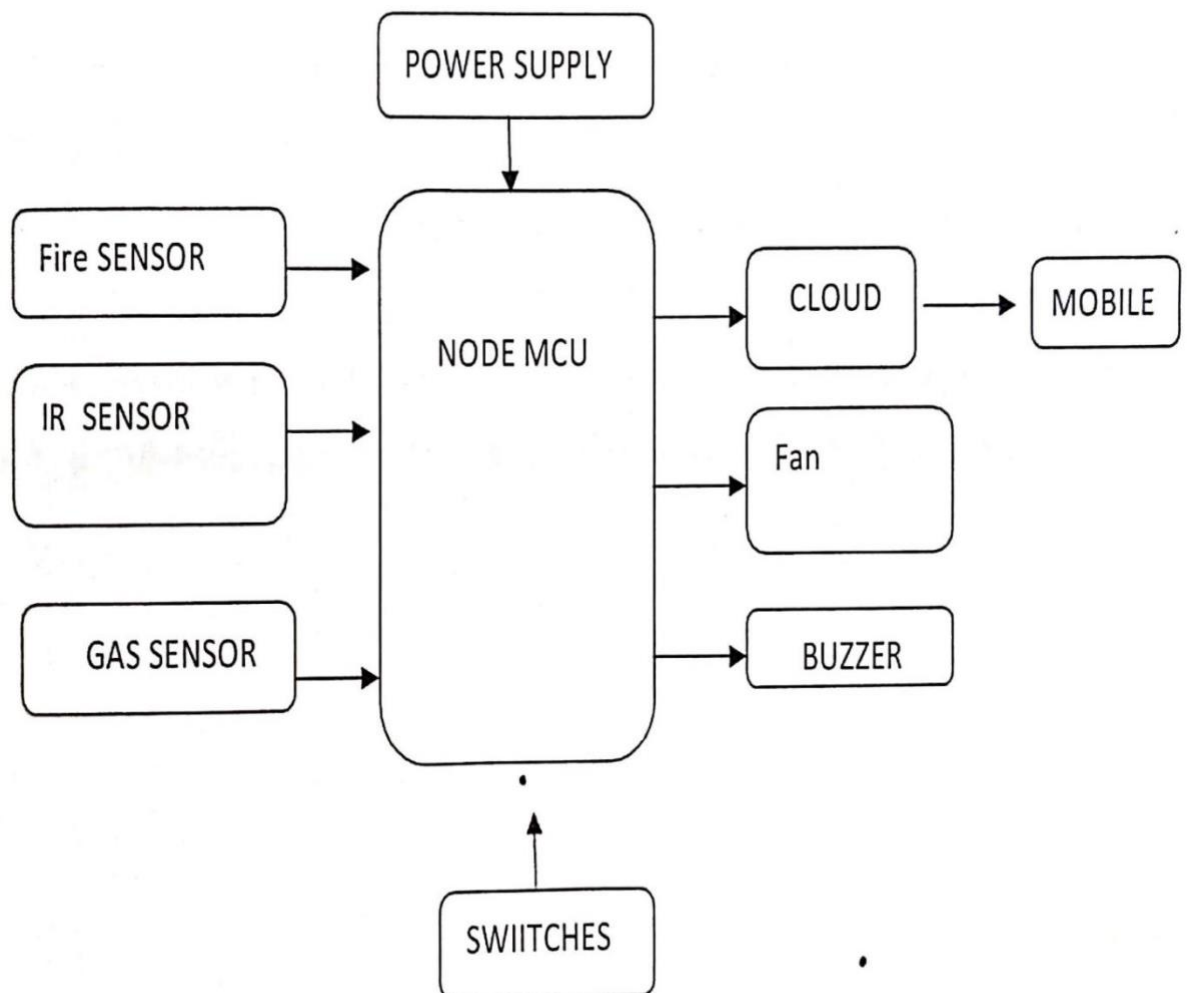
2.3 USED COMPONENTS

- NODE MCU (ESP8266)
- Flame sensor
- 2 channel Relay module
- Ultrasonic sensor
- MQ2 Sensor
- PIR Sensor
- LCD Display
- I2C module
- Buzzer
- Breadboard
- Jumper wires
- DHT11 Sensor
- Motor
- LEDS

CHAPTER-3

BLOCK DIAGRAM OF HOME AUTOMATION

3.1 BLOCK DIAGRAM



3.2 EMBEDDED SYSTEM

An embedded system is a specialized computing system that is dedicated to performing specific functions or tasks within a larger system. These systems are embedded as part of a larger device or product and are designed to operate with minimal human intervention. Here are key characteristics and examples of embedded systems.

Embedded systems play a crucial role in various industries, providing intelligent control and automation in a diverse range of applications. The design and development of embedded systems require a deep understanding of the specific requirements and constraints of the targeted application.

Every home has several examples of embedded computers. Any appliance that as a digital clock, for instances, has a small embedded microcontroller that performs no other task then to display the clock. Modern cars have embedded computers on board that control such things as ignition timing

Application of Embedded systems:

Consumer Electronics:

Smart TVs, digital cameras, washing machines, and microwave ovens often contain embedded systems to control various functions and provide user interfaces.

Automotive Systems:

Embedded systems are widely used in vehicles for functions such as engine control, anti-lock braking systems (ABS), airbag systems, navigation, and entertainment systems.

Industrial Automation:

Embedded systems are integral to industrial automation for tasks like process control, monitoring, and robotics.

Medical Devices:

Devices such as pacemakers, infusion pumps, and digital thermometers incorporate embedded systems for precise control and monitoring.

Communication Systems:

Routers, modems, and network switches contain embedded systems for managing data communication.

Smart Home Devices:

Smart thermostats, security cameras, and home automation systems utilize embedded systems for controlling and monitoring household functions.



Industrial Robots



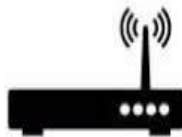
GPS Receivers



Digital Cameras



DVD Players

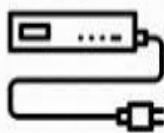


Wireless Routers

Embedded Systems



MP3 Players



Set top Boxes



Gaming Consoles



Photocopiers



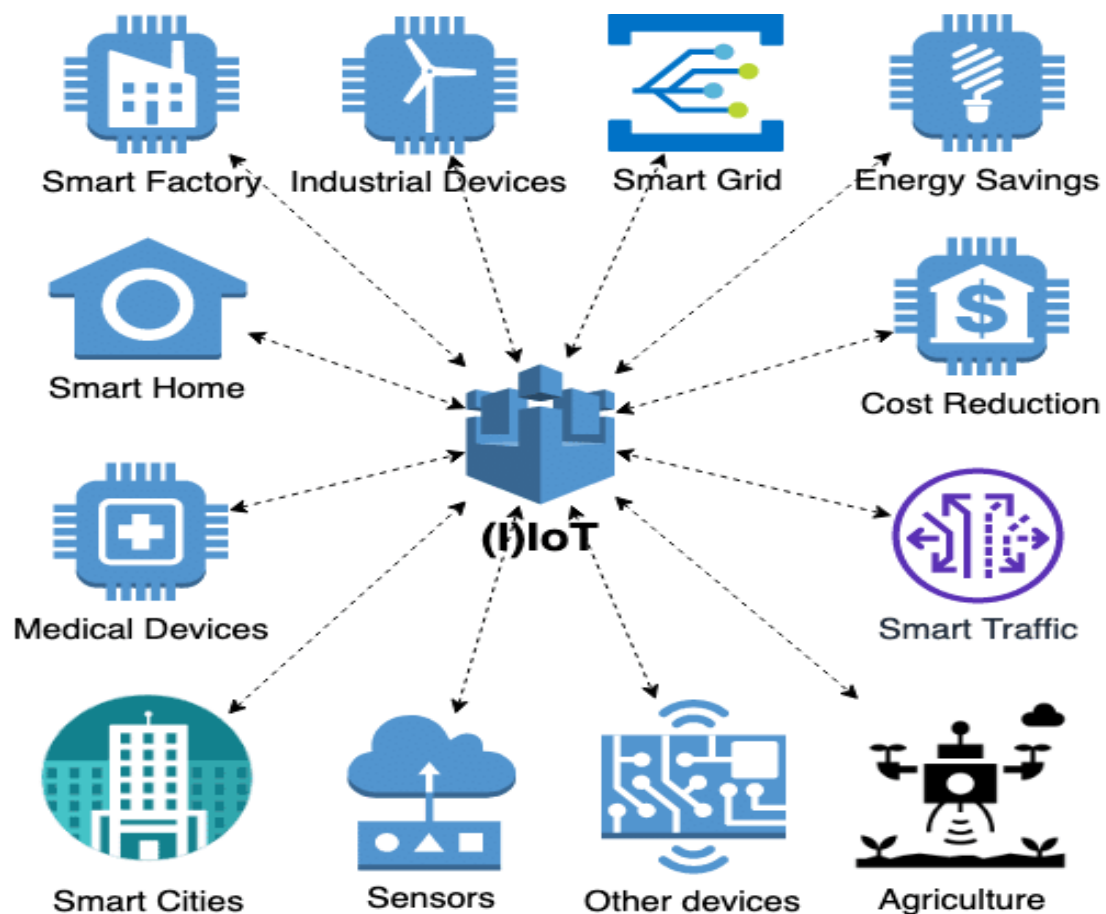
Microwave Ovens

CHAPTER-4

IOT TECHNOLOGY

4.1 About IOT

IOT as a term has evolved long way as a result of convergence of multiple technologies, machine learning, embedded systems and commodity sensors. IOT is a system of interconnected devices assigned a UIDS, enabling data transfer and control of devices over a network. It reduced the necessity of actual interaction in order to control a device. IOT is an advanced automation and analytics system which exploits networking, sensing, big data, and artificial intelligence technology to deliver complete systems for a product or service. These systems allow greater transparency, control, and performance when applied to any industry or system.



4.2 FEATURES OF IOT

Intelligence

IOT comes with the combination of algorithms and computation, software & hardware that makes it smart. Ambient intelligence in IOT enhances its capabilities which facilitate the things to respond in an intelligent way to a particular situation and supports them in carrying out specific tasks. In spite of all the popularity of smart technologies, intelligence in IOT is only concerned as a means of interaction between devices, while user and device interaction are achieved by standard input methods and graphical user interface

Connectivity

Connectivity empowers the Internet of Things by bringing together everyday objects. Connectivity of these objects is pivotal because simple object level interactions contribute towards collective intelligence in the IOT network. It enables network accessibility and compatibility in the things. With this connectivity, new market opportunities for the Internet of things can be created by the networking of smart things and applications

Dynamic Nature

The primary activity of Internet of Things is to collect data from its environment, this is achieved with the dynamic changes that take place around the devices. The state of these device change dynamically, example sleeping and waking up, connected and/or disconnected as well as the context of devices including temperature, location and speed. In addition to the state of the device, the number of devices also changes dynamically with a person, place and time

Enormous Scale

The number of devices that need to be managed and that communicate with each other will be much larger than the devices connected to the current Internet. The management of data generated from these devices and their interpretation for application purposes becomes more critical. Gartner (2015) confirms the enormous scale of IOT in the estimated report where it stated that 5.5 million new things will get connected every day and 6.4 billion connected things will be in use worldwide in 2016, which is up by 30 percent from 2015. The report also forecasts that the number of connected devices will reach 20.8 billion by 2020

Sensing

IOT wouldn't be possible without sensors that will detect or measure any changes in the environment to generate data that can report on their status or even interact with the environment. Sensing technologies provide the means to create capabilities that reflect a true awareness of the physical world and the people in it. The sensing information is simply the analog input from the physical world, but it can provide a rich understanding of our complex world

Security

IOT devices are naturally vulnerable to security threats. As we gain efficiencies, novel experiences, and other benefits from the IOT, it would be a mistake to forget about security concerns associated with it. There is a high level of transparency and privacy issues with IOT. It is important to secure the endpoints, the networks, and the data that is transferred across all of it means creating a security paradigm.

4.3 MERITS AND DEMERITS OF IOT

Merits

- Communication
- Automation and Control
- Information
- Monitor
- Save Time
- Efficient
- Improved security
- Useful for safety concern
- Improve the productivity

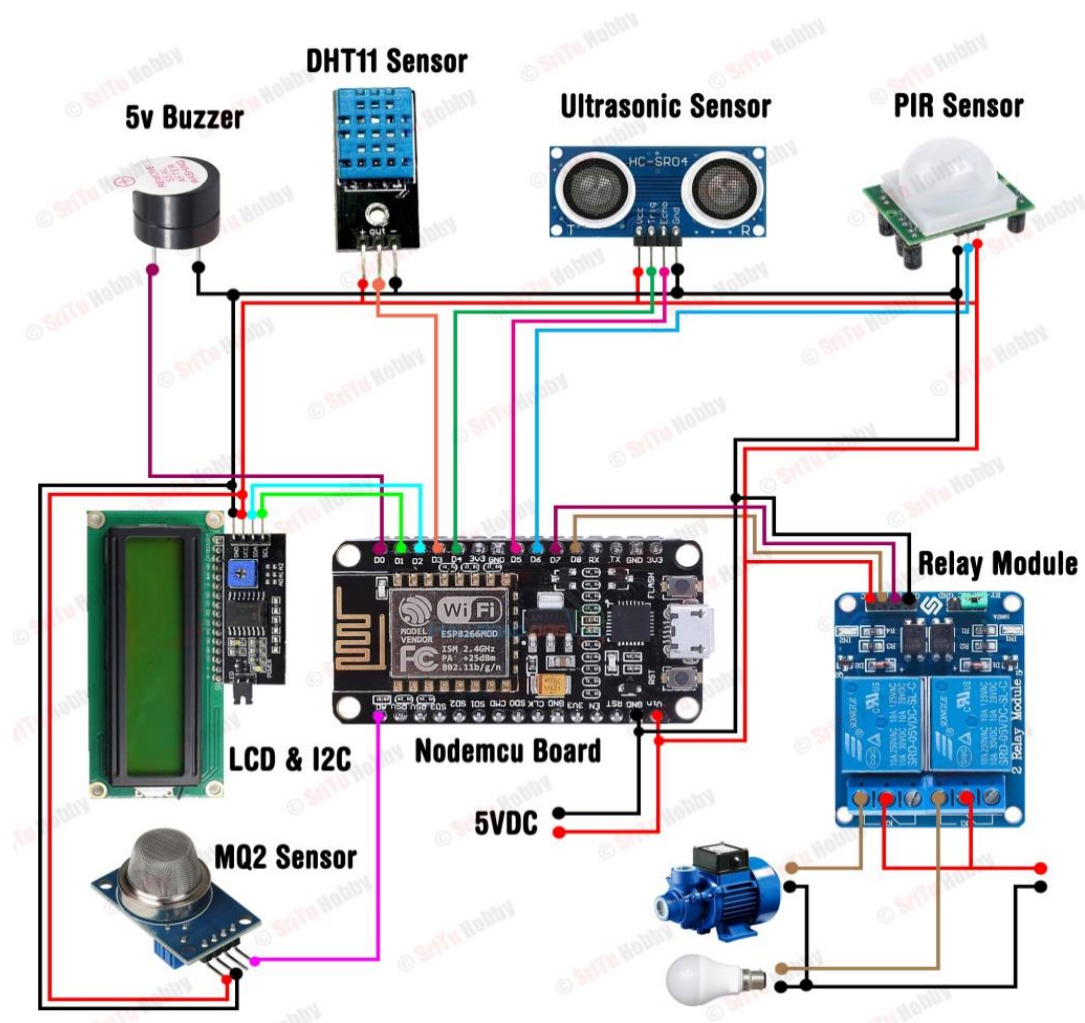
Demerits

- Compatibility
- Complexity
- Privacy/security
- Safety
- Less employment
- Highly dependent on internet

CHAPTER-5

CIRCUIT DIAGRAM AND HARDWARE IMPLEMENTATION

5.1. Circuit Diagram



5.2. Hardware Implementation

5.2.1 NODE MCU (ESP8266)

INTRODUCTION

ESP8266 is a very low-cost & user-friendly WiFi module, which develops a simple TCP/IP connection and can easily be interfaced with microcontrollers via Serial Port. The first chip in this series was ESP-01 which gained sheer attention in the market. In this tutorial, we will discuss the ESP8266 WiFi module along with its pinout, features, specifications, applications and datasheet. Let's dive in and nail down everything related to this device

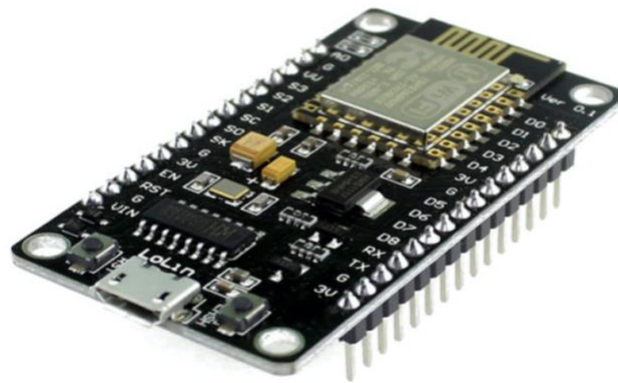
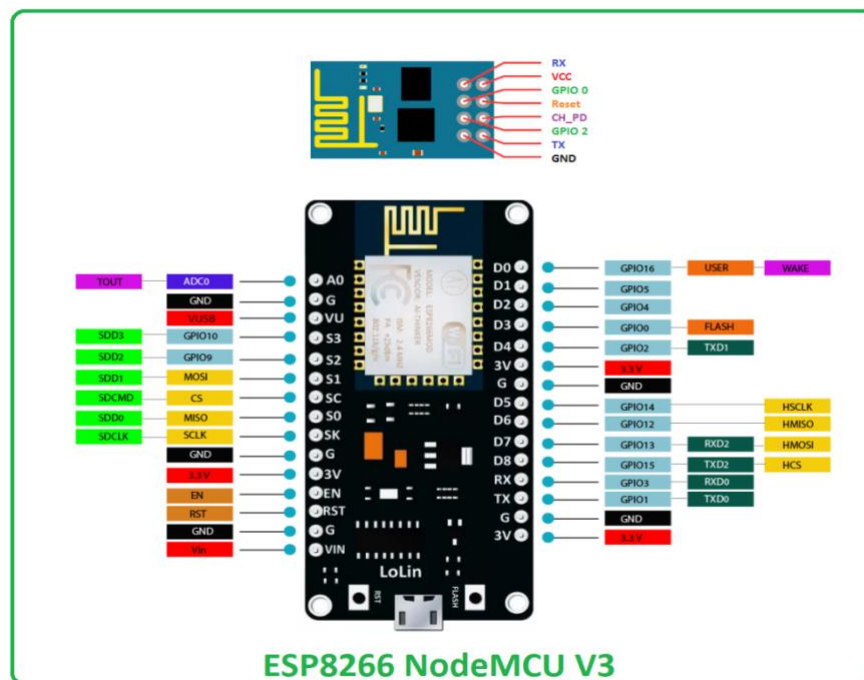


Fig: NODE MCU

ESP8266 Wi Fi Module

- **ESP8266** (also called ESP8266 Wireless Transceiver) is a cost-effective, easy-to-operate, compact-sized & low-powered **WiFi module**, designed by **Espressif Systems**, that supports both TCP/IP and Serial Protocol.
- It's normally used in IOT cloud-based embedded projects and is considered the most widely used WiFi module because of its low cost and small size.
- It runs at an operating voltage of 3V and can handle a maximum voltage of around 3.6 V, so an external logic level converter is required if you are using 5V supply.
-
-

- ESP8266 WiFi module can easily be interfaced with microcontrollers board (i.e. Arduino UNO) via Serial Port.
- There are numerous breakout boards available based on ESP8266 WiFi Module (i.e. ESP8266 Node MCU V3).
- Because of its compact size, its mostly used in autonomous projects (i.e. Robotics).



ESP8266 Pinout

- **ESP8266 Pinout** consists of 8 pins in total, which are given in below table along with their operation:

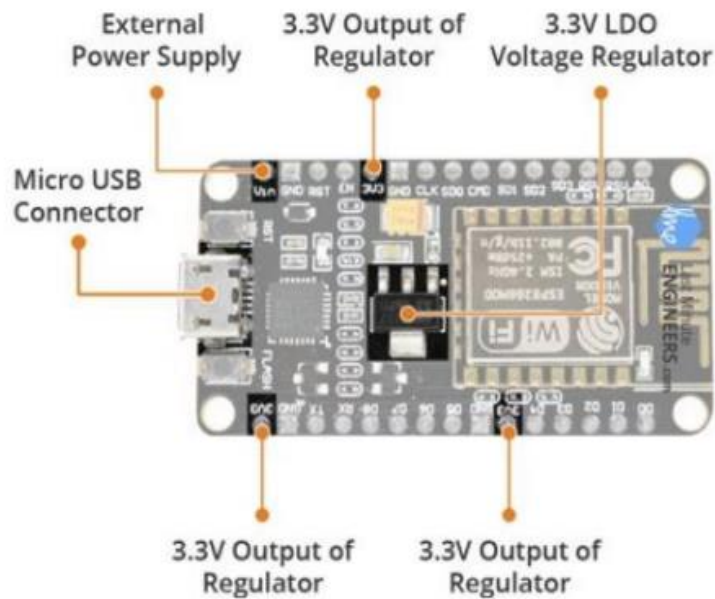
No.	Pin Name	Working
1	RX	Serial Receiver Pin
2	Vcc	Power Pin (+3.3 V; can handle up to 3.6 V)

3	GPIO 0	General-purpose I/O No. 0
4	RST	Reset
5	CH_PD	Chip power-down
6	GPIO 2	General-purpose I/O No. 2
7	TX	Serial Transmitter Pin
8	GND	Ground

ESP8266 Features

- It is also known as a system-on-chip (SoC) and comes with a 32-bit Tensilica microcontroller, antenna switches, RF balun, power amplifier, standard digital peripheral interfaces, low noise receive amplifier, power management module and filter capability.
- The processor is based on Tensilica Xtensa Diamond Standard 106Micro and runs at 80 MHz.
- It incorporates 64 KiB boot ROM, 80 KiB user data RAM and 32 KiB instruction RAM.
- It supports Wi-Fi 802.11 b/g/n around 2.4 GHz and other features including 16 GPIO, Inter-Integrated Circuit (I²C), Serial Peripheral Interface (SPI), 10-bit ADC, and I²S interfaces with DMA.
- External QSPI flash memory is accessed through SPI and supports up to 16 MiB and 512 KiB to 4 MiB is initially included in the module.
- It is a major development in terms of wireless communication with little circuitry. and contains onboard regulator that helps in providing 3.3V consistent power to the board.
- It supports APSD which makes it an ideal choice for VoIP applications and Bluetooth interfaces.

Fig: Power module on a Node MCU development board



- Operating voltage 2.5V to 3.6V
- On-board 3.6V 600mA regulator
- 80 mA operating current
- 20 μ A during sleep mode

Power Requirements

As the operating voltage range of ESP8266 is 3V to 3.6V, the board comes with a LDO voltage regulator to keep the voltage steady at 3.3V. It can reliably supply up to 600mA, which should more than enough when ESP8266 pulls as much as 80mA during RF transmissions. The output of the regulator is also broken out to one of the sides of the board and labeled as 3V3. This pin can be used to supply power to external components. Power to the ESP8266 Node MCU is supplied via the on-board Micro B USB connector.

5.2.2 GAS SENSOR

Gas sensors are essential components that are used in a variety of industries and applications to detect and measure different gases in the environment.

These sensors are essential for managing industrial operations, monitoring air quality, and guaranteeing worker safety.

It is crucial for experts and researchers interested in environmental monitoring, industrial safety, and process optimization to comprehend the capabilities and usefulness of gas sensors.



5.2.3 ULTRASONIC SENSOR

This is the HC-SR04 ultrasonic distance sensor. This economical sensor provides 2cm to 400cm of non-contact measurement functionality with a ranging accuracy that can reach up to 3mm. Each HC-SR04 module includes an ultrasonic transmitter, a receiver and a control circuit.

There are only four pins that you need to worry about on the HC-SR04: VCC (Power), Trig (Trigger), Echo (Receive), and GND (Ground).



FIG: ULTRASONIC SENSOR

5.2.4 DHT11 Sensor

The **DHT11** is a commonly used **Temperature and humidity sensor**. The sensor comes with a dedicated NTC to measure temperature and an 8-bit microcontroller to output the values of temperature and humidity as serial data. The sensor is also factory calibrated and hence easy to interface with other microcontrollers.

The sensor can measure temperature from 0°C to 50°C and humidity from 20% to 90% with an accuracy of $\pm 1^\circ\text{C}$ and $\pm 1\%$. So if you are looking to measure in this range then this sensor might be the right choice for you.

5.2.5 PIR Sensor

The term PIR is the short form of the Passive InfraRed. The term “passive” indicates that the sensor does not actively take part in the process, which means, it does not emit the referred IR signals itself, rather passively detects the infrared radiations coming from the human body in the surrounding area.

The Passive infrared sensors consist of three pins as indicated in the diagram shown above.

- Pin1 corresponds to the drain terminal of the device, which should be connected to the positive supply 5V DC.
- Pin2 corresponds to the source terminal of the device, which should be connected to the ground terminal via a 100K or 47K resistor. The Pin2 is the output pin of the sensor, and the detected IR signal is carried forward to an amplifier from the pin 2 of the sensor.
- Pin3 of the sensor is connected to the ground.

5.2.6 Relay Module

Relay modules (or power relay modules) are ubiquitous electronic components. They are an exceedingly significant component of any home automation project. You will require a relay module if you use a low voltage microcontroller such as an Arduino to control motors or lighting circuits.

Relay modules are straightforward components. Essentially, they work as switches. Your average relay module comprises two internal metal contacts. Usually, these contacts do not connect or touch each other. However, relays include an internal switch connecting these contacts to complete an electrical circuit that allows current flow.

Relay modules do not work like manual light switches. To illustrate, when you switch on a light, you must press a button to connect the two metal contacts within it. Inversely, a relay switch uses electric pulses to turn its internal switch on and off

5.2.7 LCD Display

LCD (Liquid Crystal Display) is a type of flat panel display which uses liquid crystals in its primary form of operation. LEDs have a large and varying set of use cases for consumers and businesses, as they can be commonly found in smartphones, televisions, computer monitors and instrument panels.

LCDs were a big leap in terms of the technology they replaced, which include light-emitting diode (LED) and gas-plasma displays. LCDs allowed displays to be much thinner than cathode ray tube (CRT) technology. LCDs consume much less power than LED and gas-display displays because they work on the principle of blocking light rather than emitting it. Where an LED emits light, the liquid crystals in an LCD produces an image using a backlight.

As LCDs have replaced older display technologies, LCDs have begun being replaced by new display technologies such as OLEDs.

CHAPTER-6

SOFTWARE IMPLEMENTATION

6.1 Software Tools Required

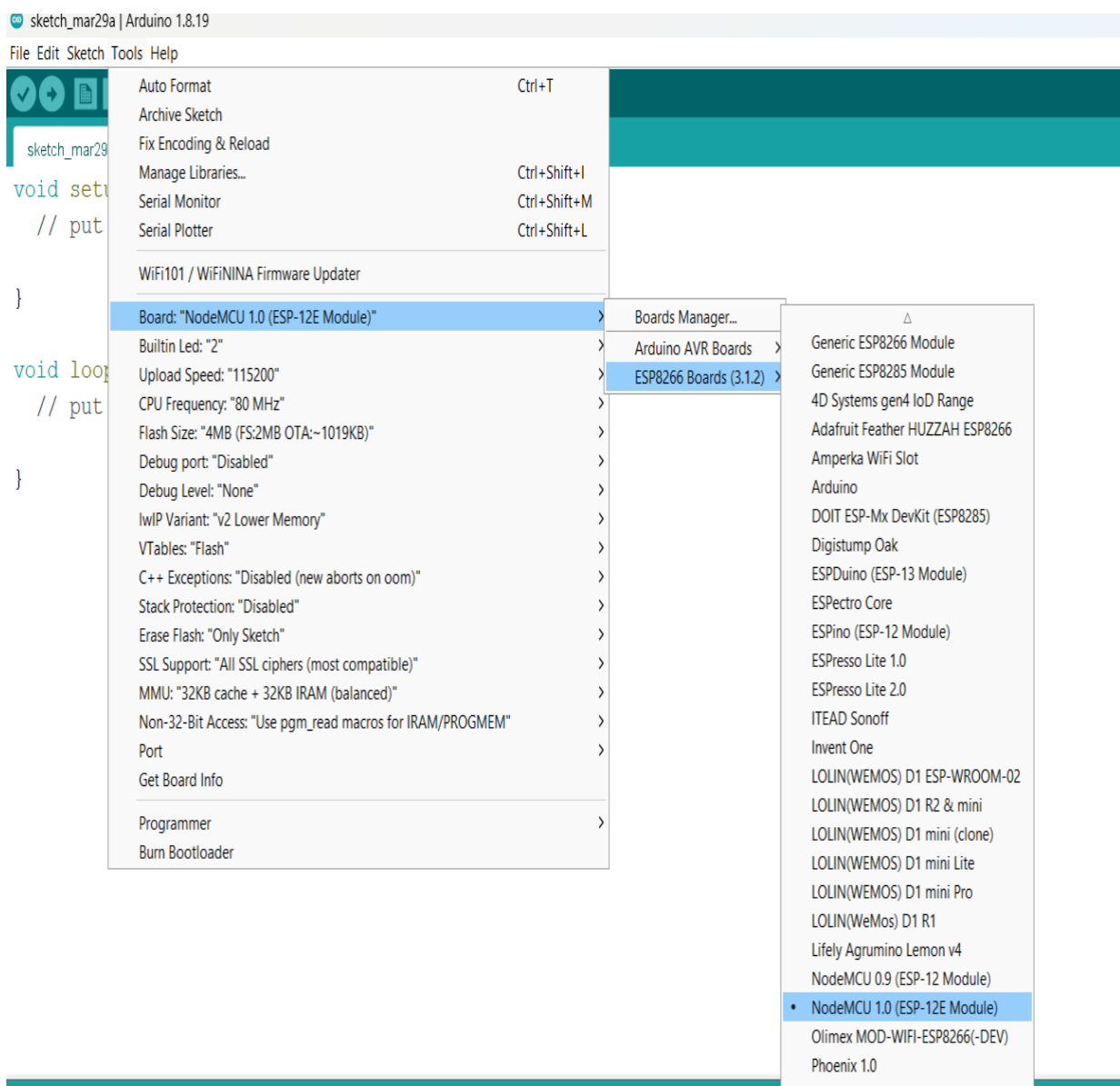
Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. You can tell your board what to do by sending a set of instructions to the microcontroller on the board. To do so you use the Arduino programming language (based on Wiring), and the Arduino Software (IDE), based on Processing

Over the years Arduino has been the brain of thousands of projects, from everyday objects to complex scientific instruments. A worldwide community of makers - students, hobbyists, artists, programmers, and professionals - has gathered around this open-source platform, their contributions have added up to an incredible amount of accessible knowledge that can be of great help to novices and experts alike. Arduino was born at the Ivrea Interaction Design Institute as an easy tool for fast prototyping, aimed at students without background in electronics and programming. As soon as it reached a wider community, the Arduino board started changing to adapt to new needs and challenges, differentiating its offer from simple 8-bit boards to products for IoT applications, wearable, 3D printing, and embedded environments. All Arduino boards are completely open-source, empowering users to build them independently and eventually adapt them to their particular needs.

6.2 Arduino IDE Board setup Procedure

If you downloaded the Arduino IDE before plugging in your Arduino board, when you plugged in the board, the USB drivers should have installed automatically. The most recent Arduino IDE should recognize connected boards and label them with which COM port they are using. Select the Tools pulldown menu and then Port. Here it should list all open COM ports, and if there is a recognized Arduino Board, it will also give it's name. Select the Arduino board that you have connected to the PC. If the setup was successful, in the bottom right of the Arduino IDE, you should see the board type and COM number of the board you plan to program.

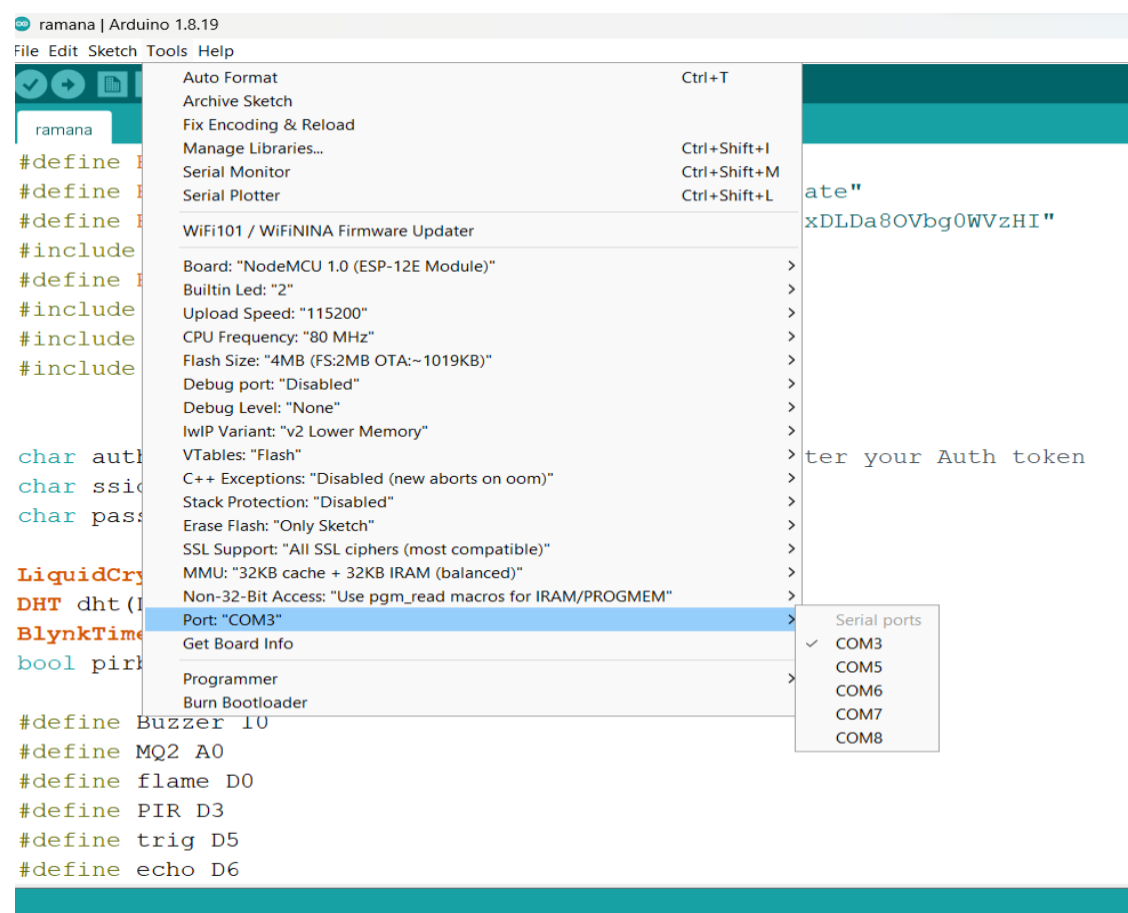
Note: the Arduino Uno occupies the next available COM port; it will not always be COM3.



Arduino IDE: COM SETUP

At this point, your board should be set up for programming, and you can begin writing and uploading code. One common procedure to test whether the board you are using is properly set up is to upload the "Blink" sketch. This sketch is included with all Arduino IDE releases and can be accessed by the File pull-down menu and going to Examples Basics, and then select Blink. Standard Arduino Boards include a surface-mounted LED labeled "L" or "LED" next to the "RX" and "TX" LEDs, that is connected to digital pin

. This sketch will blink the LED at a regular interval, and is an easy way to confirm if your board is set up properly and you were successful in uploading code. Open the "Blink" sketch and press the "Upload" button in the upper-left corner to upload "Blink" to the board



6.3 Project Coding

```
#define BLYNK_TEMPLATE_ID      "TMPL3DJFG3nM7"
#define BLYNK_TEMPLATE_NAME    "Quickstart Template"
#define BLYNK_AUTH_TOKEN      "xK913XiWw5ihlcZTxDLDa8OVbg0WVzHI"
#include <LiquidCrystal_I2C.h>
#define BLYNK_PRINT Serial
#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>
#include <DHT.h>

char auth[] = "xK913XiWw5ihlcZTxDLDa8OVbg0WVzHI";//Enter your Auth token
char ssid[] = "Ramana";//Enter your WIFI name
char pass[] = "12345679";//Enter your WIFI password

LiquidCrystal_I2C lcd(0x27, 16, 2);
DHT dht(D4, DHT11); //(sensor pin,sensor type)
BlynkTimer timer;
bool pirbutton = 0;

#define Buzzer 10
#define MQ2 A0
#define flame D0
#define PIR D3
#define trig D5
#define echo D6
```



```

#define relay1 D7
#define relay2 D8

BLYNK_WRITE(V0) {
    pirbutton = param.asInt();
}

void setup() {
    Serial.begin(9600);
    lcd.init();
    lcd.backlight();
    pinMode(Buzzer, OUTPUT);
    pinMode(flame, INPUT);
    pinMode(PIR, INPUT);
    pinMode(trig, OUTPUT);
    pinMode(echo, INPUT);
    pinMode(relay1, OUTPUT);
    pinMode(relay2, OUTPUT);
    digitalWrite(relay1, HIGH);
    digitalWrite(relay2, HIGH);
    Blynk.begin(auth, ssid, pass);
    dht.begin();
    timer.setInterval(100L, gassensor);
    timer.setInterval(100L, DHT11sensor);
    timer.setInterval(100L, flamesensor);
    timer.setInterval(100L, pirsensor);
    timer.setInterval(100L, ultrasonic);
}

```

```

void gassensor() {
  int value = analogRead(MQ2);
  Serial.println(value);
  value = map(value, 0, 1024, 0, 100);
  if (value <= 35) {
    digitalWrite(Buzzer, LOW);
  } else if (value > 35) {
    Blynk.notify("Warning! Gas leak detected");
    digitalWrite(Buzzer, HIGH);
  }
  Blynk.virtualWrite(V1, value);
  lcd.setCursor(9, 0);
  lcd.print("G :");
  lcd.print(value);
  lcd.print(" ");
}

void DHT11sensor() {
  float h = dht.readHumidity();
  float t = dht.readTemperature();

  if (isnan(h) || isnan(t)) {
    Serial.println("Failed to read from DHT sensor!");
    return;
  }

  Blynk.virtualWrite(V2, t);
  Blynk.virtualWrite(V3, h);
}

```

```

lcd.setCursor(0, 0);
lcd.print("T :");
lcd.print(t);

lcd.setCursor(0, 1);
lcd.print("H :");
lcd.print(h);
}

void flamesensor() {
  bool value = digitalRead(flame );
  if (value == 1) {
    digitalWrite(Buzzer, LOW);
  } else if (value == 0) {
    Blynk.notify("Warning! Fire was detected");
    digitalWrite(Buzzer, HIGH);
  }
}

void pirsensor() {
  bool value = digitalRead(PIR);
  if (pirbutton == 1) {
    if (value == 0) {
      digitalWrite(Buzzer, LOW);
    } else if (value == 1) {
      Blynk.notify("Warning! Please check your security system");
      digitalWrite(Buzzer, HIGH);
    }
  }
}
}

```

```
void ultrasonic() {  
    digitalWrite(trig, LOW);  
    delayMicroseconds(4);  
    digitalWrite(trig, HIGH);  
    delayMicroseconds(10);  
    digitalWrite(trig, LOW);  
    long t = pulseIn(echo, HIGH);  
    long cm = t / 29 / 2;  
    Blynk.virtualWrite(V4, cm);  
    lcd.setCursor(9, 1);  
    lcd.print("W :");  
    lcd.print(cm);  
    lcd.print(" ");  
}
```

```
void loop() {  
    Blynk.run();  
    timer.run();  
}
```

CHAPTER-7

ADVANTAGES AND APPLICATIONS

7.1 Advantages

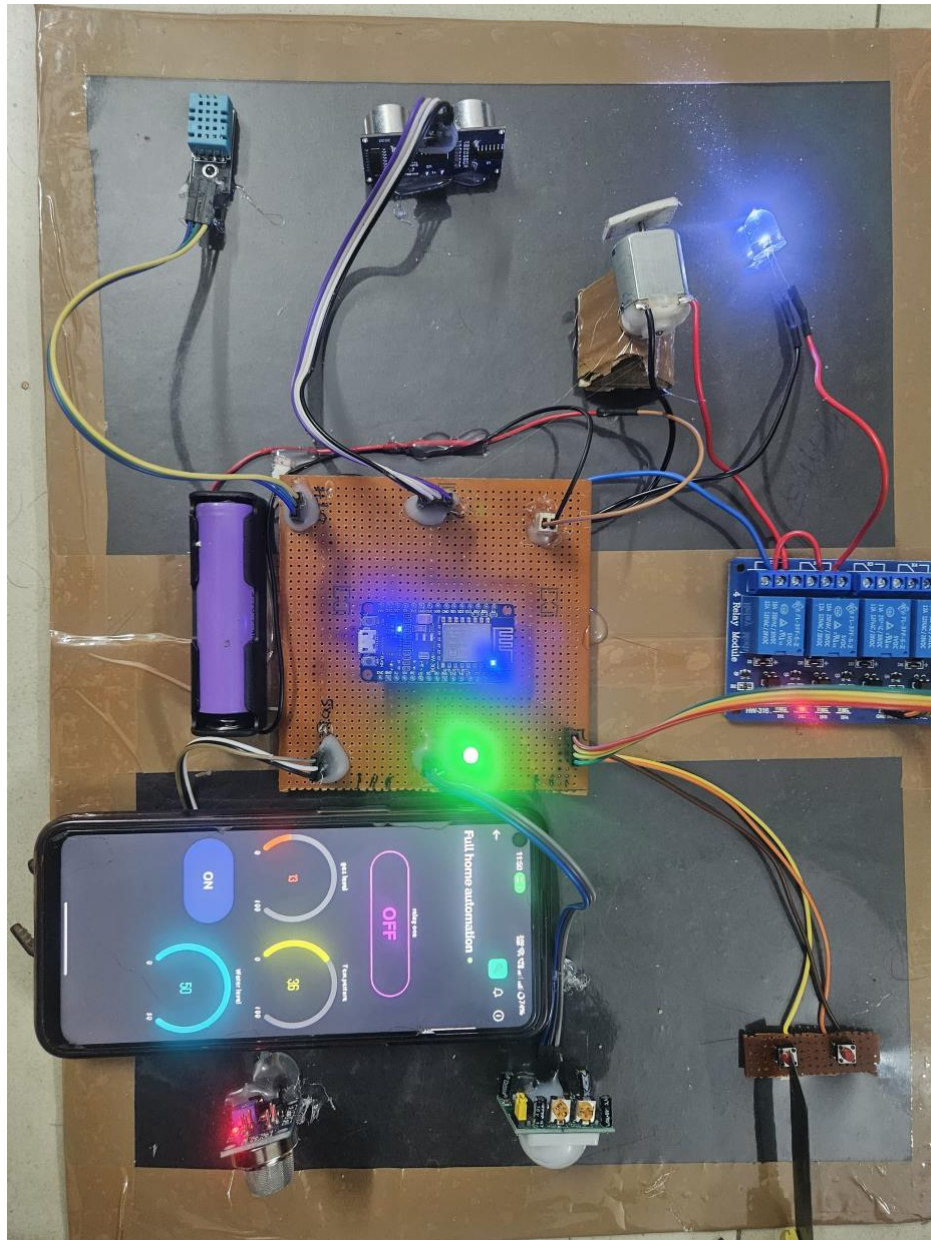
- Convenience
- Energy efficiency
- Safety and security
- Save time
- Remote accessibility
- Flexibility
- Increase home comfort
- Enhance accessibility

7.2 Applications

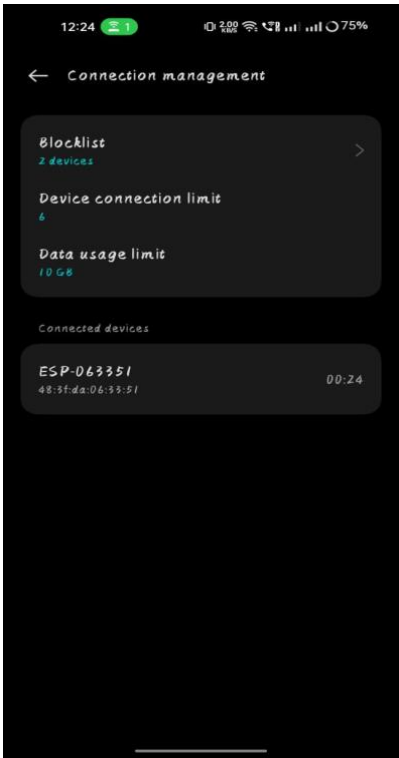
- Security and Surveillance
- Energy management
- Health and Well being
- Leak detection and Prevention
- Lighting
- Water management
- Remote Monitoring and Controls

CHAPTER-8

OUTPUT



WiFi Connection



BLYNK INTERFACE



CONCLUSION

Home automation offers unparalleled convenience, comfort, security, and efficiency. By integrating smart devices and systems, homeowners can streamline daily tasks, enhance safety, and reduce energy consumption. As the technology evolves, the potential for customization and improvement in quality of life continues to expand, making home automation an integral part of modern living. Home automation revolutionizes daily life by offering seamless control and monitoring of various household systems. With smart devices and sensors, homeowners can enjoy enhanced convenience, security, and energy efficiency. As the technology advances, the possibilities for customization and integration into daily routines are limitless, making smart homes the standard for modern living.

REFERENCE

Websites

<https://www.alldatasheet.com/datasheet-pdf/pdf/1132995/ESPRESSIF/ESP8266.html>

<https://justdoelectronics.com/home-automation-using-esp8266-blynk2-0-app/#Introduction>